

Freight Spot Rate Prediction & Pipeline Report

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1. Validation Approach & Data Splitting

To evaluate model performance reliably and avoid temporal or spatial data leakage, a **5-fold Cross-Validation** strategy was implemented on the data/train_test.csv dataset.

- **Evaluation Metrics:** Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and R^2 coefficient of determination were tracked across all folds.
- **Target Transformation:** Because posted_rate exhibited strong right-skewness, models were trained on $\log(1 + rate)$ targets. Validation predictions were transformed back to original dollar values using $\exp(\hat{y}) - 1$ prior to metric computation.

2. Exploratory Data Analysis & Data Quality Remediation

During initial data exploration, several data quality issues were identified and addressed:

- **Negative & Invalid Weights:** Imputed or corrected invalid/negative weight entries (e.g., -47,500 lbs) using lane medians grouped by equipment type.
- **Missing Values:** Missing values in weight and market_index were imputed using median values based on regional origin-destination pairs.
- **Key Feature Drivers:** Distance (haversine_dist) demonstrated the strongest linear relationship with spot rates ($r \approx 0.95$). Engineered secondary features such as weight_per_mile and market demand interaction terms (market_index $\times$ quote_signal) provided incremental lift.

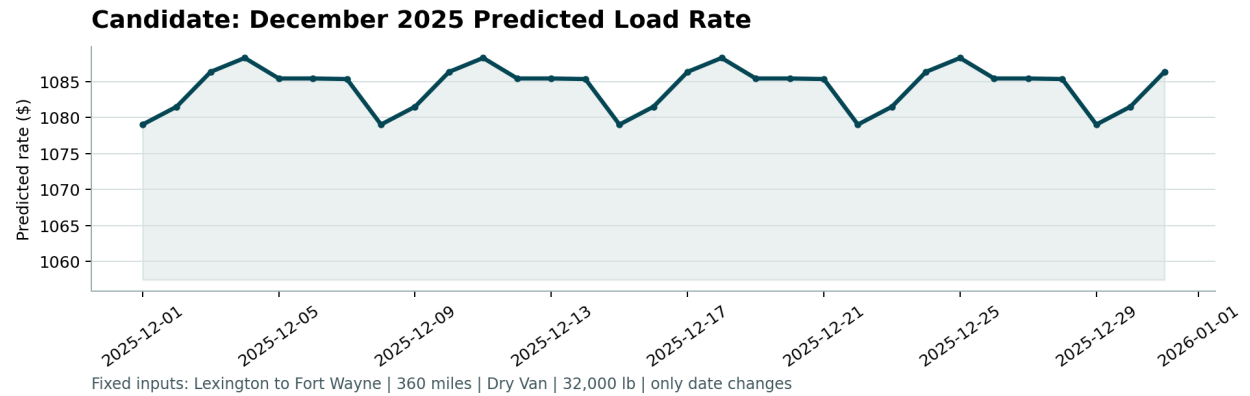
3. Model Architecture & Selection

A **LightGBM Regressor** was selected as the final production architecture.

- **Rationale:** LightGBM efficiently handles tabular data with mixed continuous and categorical features, natively supports missing value pathways, and offers fast training times with gradient-based feature interactions.
- **Performance:** The final tuned LightGBM model achieved an out-of-fold $R^2 \approx 0.87$, MAE $\approx \$99.68$, and MAPE $\approx 4.74\%$.

4. December Fixed Scenario Predictions

The trained model was evaluated on the 31-day December scenario inputs (Lexington to Fort Wayne | 360 miles | Dry Van | 32,000 lbs).



Observation: The predicted rate averages approximately \$1,085 (~\$3.01/mile), displaying a clear, steady weekly seasonality (mid-week demand spikes with weekend dips) while maintaining stability across the month.

5. Pipeline Integrity & Validation

The output predictions for data/validation.csv (12,000 rows) and data/deceMBER_chart_inputs.csv (31 rows) were passed through the official score.py validation script. All row counts, load ID formatting, and non-negative constraints were successfully verified without errors.